Practical No.: 8

Title: Demonstrating the Use of POST Method in ReactJS & NodeJS

Aim:

To demonstrate how to **send data from a React frontend to a Node.js backend** using the **POST method** with the `fetch` API.

Theory:

What is the POST Method?

The **POST method** is used to **send data** to the server. It is commonly used in **form submissions** and **API interactions** to create or update resources.

Steps Involved:

- Create a **React form** to accept user input.
- Use the **fetch API** to send the data to the Node.js backend via **POST request**.
- Create a **Node.js server using Express** to handle the request and return a response.

Syntax of POST Request in React:

```
fetch("http://localhost:5000/submit", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify({ name: "John" }),
})
.then(response => response.json())
.then(data => console.log(data));
```

ReactJS Code

Step 1: Create a React Form (index.html & post_request.js)

HTML File (index.html)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>React POST Method</title>
    <script src="https://unpkg.com/react@18/umd/react.development.js"></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-
dom.development.js"></script>
    <script src="https://unpkg.com/babel-standalone@6/babel.min.js"></script>
</head>
<body>
    <h2>Practical 8: POST Method in React</h2>
    <div id="root"></div>

    <script type="text/babel" src="post_request.js"></script>
```

```
</body>
</html>
```

ReactJS File (post_request.js)

```
function App() {
    const [name, setName] = React.useState("");

    const handleSubmit = (e) => {
        e.preventDefault();
        fetch("http://localhost:5000/submit", {
            method: "POST",
            headers: { "Content-Type": "application/json" },
            body: JSON.stringify({ name }),
        })
        .then(response => response.json())
        .then(data => alert("Response from Server: " + data.message));
    };

    return (
        <div>
            <h3>Enter Your Name</h3>
            <form onSubmit={handleSubmit}>
                <input
                    type="text"
                    value={name}
                    onChange={(e) => setName(e.target.value)}
                    placeholder="Enter name"
                    required
                />
                <button type="submit">Submit</button>
            </form>
        </div>
    );
}

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<App />);
```

Step 2: Create a Node.js Backend (server.js)

```
const express = require("express");
const cors = require("cors");
const app = express();

app.use(express.json()); // Parse JSON requests
app.use(cors()); // Enable CORS for frontend-backend communication

app.post("/submit", (req, res) => {
    const { name } = req.body;
    res.json({ message: `Hello, ${name}! Your data has been received.` });
});

app.listen(5000, () => console.log("Server running on port 5000"));
```

Steps to Run the Application:

1. **Start the Backend:**
   - Install dependencies: `npm install express cors`
   - Run the server: `node server.js`
2. **Open `index.html` in the browser** and enter a name.
3. Click the **Submit** button → It sends the data to the backend.
4. The server responds with **"Hello, {name}! Your data has been received."**